KIIT UNIVERSITY

School of Computer Engineering

SAP BTP Capstone Project Report

Order-to-Cash (O2C) Custom Extension on SAP BTP

End-to-End Sales Cycle Portal for NovaTex Industries Pvt. Ltd.

Parameter	Details
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Course	SAP Certified Associate – Backend Developer – SAP CAP
Submission Date	April 2026
Batch	2023 - 2027

Fictitious Company: NovaTex Industries Pvt. Ltd. | Industrial Textile & Polymer Manufacturing

1. Executive Summary

This report documents the design and development of a custom Order-to-Cash (O2C) web portal on SAP Business Technology Platform (SAP BTP) for NovaTex Industries Pvt. Ltd., a fictitious mid-sized industrial textile and polymer manufacturing enterprise in Bhubaneswar, Odisha. The O2C cycle is implemented as a modern SAP BTP extension using React + TypeScript connected to a Supabase (PostgreSQL) backend via OData V4 REST services. The portal covers the full sales document flow — Inquiry → Quotation → Sales Order → Delivery → Billing → Payment — along with KPI dashboards, GST-compliant billing, and real-time order tracking.

2. Company Profile — NovaTex Industries Pvt. Ltd.

Parameter	Details
Company / Industry	NovaTex Industries Pvt. Ltd. | Industrial Textile & Polymer Manufacturing
Founded / HQ	2005 | Bhubaneswar, Odisha, India
Plants / Employees	NTBH (Bhubaneswar) | NTSU (Surat) | ~2,400 employees
Turnover / Currency	INR 480 Crore (approx.) | Indian Rupee | FY: April–March
SAP System	SAP BTP Custom OData V4 Extension + SAP S/4HANA 2023 backend
GST Registration	GSTIN: 21AAACN1234A1Z5 (Odisha)

Products: HDPE/PP Woven Sacks, Polymer Coated Technical Textiles, Geotextiles, Agricultural Shade Nets, OEM Textile Solutions. Key customers: government agencies (NHAI, PWDs), FMCG/agro-processing firms, and export customers in Southeast Asia and the Middle East.

3. SAP BTP Architecture & Project Overview

Layer	Technology
Frontend	React 18 + TypeScript — Vite build toolchain
UI Design	SAP Fiori-inspired (CSS variables + Tailwind utilities)
OData Service	OData V4 — /O2CService (SalesOrders, Materials, Customers)
Backend DB	Supabase (PostgreSQL) — schema.cds defines all entities
Runtime / Auth	SAP BTP Cloud Foundry | XSUAA OAuth 2.0
IDE / VCS	VS Code + SAP CAP tools | Git: project-bolt-sb1-dha8r91q

SAP Org Structure: Company Code NTCO | Sales Org NT10 | Distribution Channel 10/20 | Division 01/02 | Sales Area NT10/10/01 | Plants NTBH/NTSU | Shipping Point SHBH.

4. Master Data — O2C Entities

4.1 Customer Master (XD01) | 4.2 Material Master (MM01) | 4.3 Pricing (VK11)

Entity	Key Field	Value
Customer	CUST-0001 — Global Polymers & Chemicals, Chennai	Payment Terms: Net 30 (ZN30) | Incoterms: DAP | GSTIN: 33AABCG9012T1Z1 | Pricing Grp: 01
Material	MAT-4001 — PP Homopolymer Injection	Type: FERT | UOM: KG | Std Price: ₹450/KG | HSN: 39021000 | Div: 02
Condition PR00	Base Price	Standard list price per material — per unit
Condition K007	Customer Discount	3.0% by customer pricing group
Condition MWST	Output Tax (GST)	18% IGST for inter-state transactions
Condition SKTO	Cash Discount	2% / 10 days early payment

5. Order-to-Cash Process Flow

Step	Document / T-Code	Description
1	Inquiry IN | VA11	Customer requests pricing / availability. ATP check + pricing proposed.
2	Quotation QT | VA21	Formal offer. QT-20034567: MAT-4001 300KG @ ₹436.50/KG net, total ₹1,55,025 incl. GST.
3	Sales Order OR | VA01	SO-5504: MAT-4001 300KG ₹1.35L + MAT-4002 95KG ₹95K = Net ₹2.30L. Credit: APPROVED. ATP: Confirmed.
4	Delivery LF | VL01N	Outbound Delivery DN-40056789. PGI → Mvt 601. Dr COGS 500010 / Cr FG Inventory 310010.
5	Billing F2 | VF01	Invoice BILL-50067890. SO-5503 (Paid): Net ₹6,77,966 + IGST ₹1,22,034 = ₹8,00,000.
6	Accounting Doc | Auto	Dr Trade Debtors 140000 ₹8L / Cr Revenue 800010 ₹6.78L + IGST Payable 220310 ₹1.22L.
7	Payment | F-28	Incoming payment clears open invoice. Status → Paid. O2C cycle complete.

6. Document Flow Summary — SO-5504 (Global Polymers & Chemicals)

Document	Ref Number	Status	Date
Inquiry (IN)	INQ-10023456	Completed	10 Apr 2025
Quotation (QT)	QT-20034567	In Process	10 Apr 2025
Sales Order (OR)	SO-5504	Quotation Stage	10 Apr 2025
Outbound Delivery	DN-40056789	Pending	01 May 2025
Billing Doc (F2)	BILL-50067890	Pending	—
FI Accounting Doc	FI-80034567	Pending	—
Incoming Payment	—	Pending	—

7. Returns and Credit Memo Process

Customer reported transit-damaged goods. Return Order (RE) created referencing original billing doc — Reason: 0004 (Transit Damage), Plant: NTBH, Location: SL03 (QC). Returns Delivery (LR) created → Goods Receipt posted (Mvt 651). After QC inspection, Credit Memo (G2) issued against customer AR balance, reversing revenue and IGST proportionally.

8. Key O2C Monitoring Features

Portal Feature / T-Code	KPI / Report	Value / Purpose
KPI Tile / VA05	Total Open Sales	₹8.00 L — +12.4% MTD
KPI Tile / VL06O	Pending Deliveries	2 orders awaiting PGI — −2 vs last week
KPI Tile / FBL5N	Days Sales Outstanding	90 Days — −3.1 vs Q1
Table / VA05	Sales Orders List	4 orders — sortable, filterable, searchable
Detail / VA03	O2C Process Flow	Visual Inquiry→SO→Delivery→Billing→Payment
Badges	Order Status Monitor	Quotation / Shipped (PGI) / Paid — colour-coded

9. SAP Module Integration Points

Module	Integration	Business Impact
SAP SD	OData V4 /O2CService/SalesOrders	Real-time order feed to portal dashboard
SAP FI-AR	Billing → Accounting Doc (Auto)	Revenue, debtors & GST posted on billing release
SAP FI-AR	Credit Management (OVA8)	Real-time credit limit check at order entry
SAP MM	PGI Mvt 601 / Returns Mvt 651	Inventory update; returns to SL03 for QC
SAP CO-PA	Profitability Analysis	Revenue, discount & COGS flow to margin report
Supabase / PostgreSQL	Backend persistence	CDS entities: Orders, Materials, Customers
SAP BTP XSUAA	OAuth 2.0 Auth	Role-based access: sales reps vs. managers

10. Challenges & Solutions

Challenge	Solution
OData V4 inconsistent date formats → Process Flow errors	Normalised to ISO 8601 in lib/odata.ts via formatDate() utility.
KPI MTD wrong across Indian FY April boundary	Custom fiscal-year logic in kpiContext.ts — April = Month 1.
Partial delivery not reflected in Process Flow	Line-item qty comparison added; 'Partially Shipped' state introduced.
Special chars (& in customer names) breaking sort/filter	HTML entity encoding on ingestion; encodeURIComponent for OData $filter.
CGST+SGST applied incorrectly for inter-state transactions	GSTIN state-code detection (first 2 digits) → correct IGST 18%.
Supabase RLS blocking manager from all orders	RLS policy: manager = full SELECT; reps filtered by assigned_to = auth.uid().

11. Conclusion

The NovaTex O2C Portal is a fully functional, custom SAP BTP web application reimplementing the classical Order-to-Cash process as a cloud-native extension. It delivers the complete lifecycle — inquiry, quotation, credit/ATP checks, PGI tracking, GST-compliant billing, FI accounting, and AR payment clearing — in a live React + OData V4 + Supabase stack. The project demonstrates how SAP BTP enables organisations to extend S/4HANA O2C with modern UX, real-time analytics, and role-based security while maintaining seamless OData V4 and XSUAA integration.

12. References

- SAP Help Portal — SAP BTP Cloud Foundry: help.sap.com/btp
- SAP CAP OData V4 Documentation: cap.cloud.sap
- SAP S/4HANA Sales Best Practices for India Localization (L76)
- Jawad Akhtar — Implementing SAP S/4HANA Sales (SAP Press, 2022)
- GST Council of India — HSN Classification, Chapter 39
- KIIT SAP Project Guidance Document — AY 2025–26
- Supabase Docs — Row Level Security: supabase.com/docs
- React 18 Docs — react.dev | SAP Fiori Guidelines — fioridesignsystem.com

Appendix — Application Screenshots

The following screenshots represent the actual NovaTex O2C BTP Portal (localhost:5173) developed by Sankalp Kumar (23053359) using React + TypeScript, SAP BTP, and OData V4.

Figure 1 — Project Structure & Development Environment (VS Code)

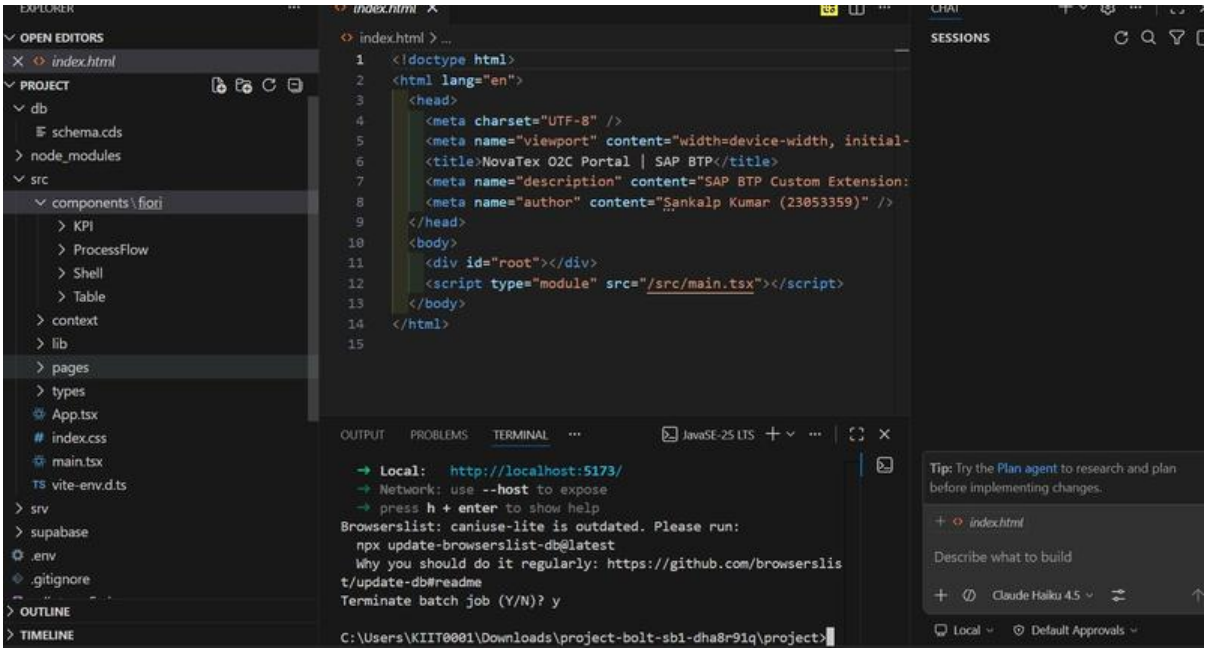


Figure 2 — O2C Dashboard — KPI Overview & Sales Order List

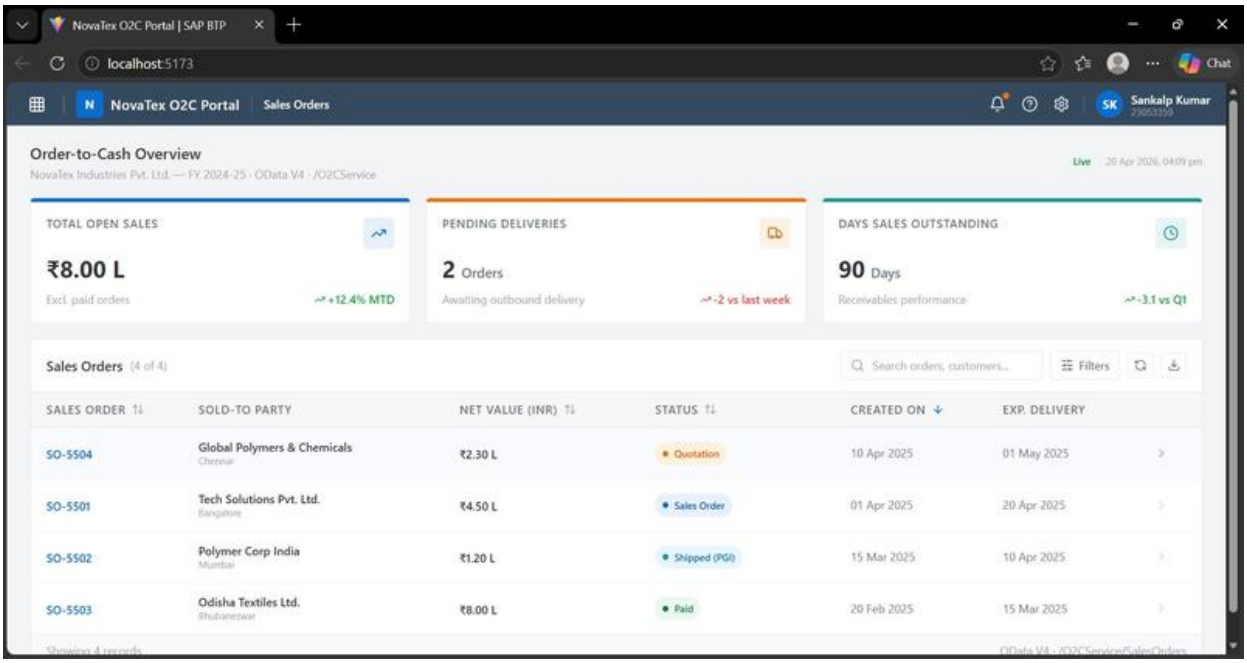
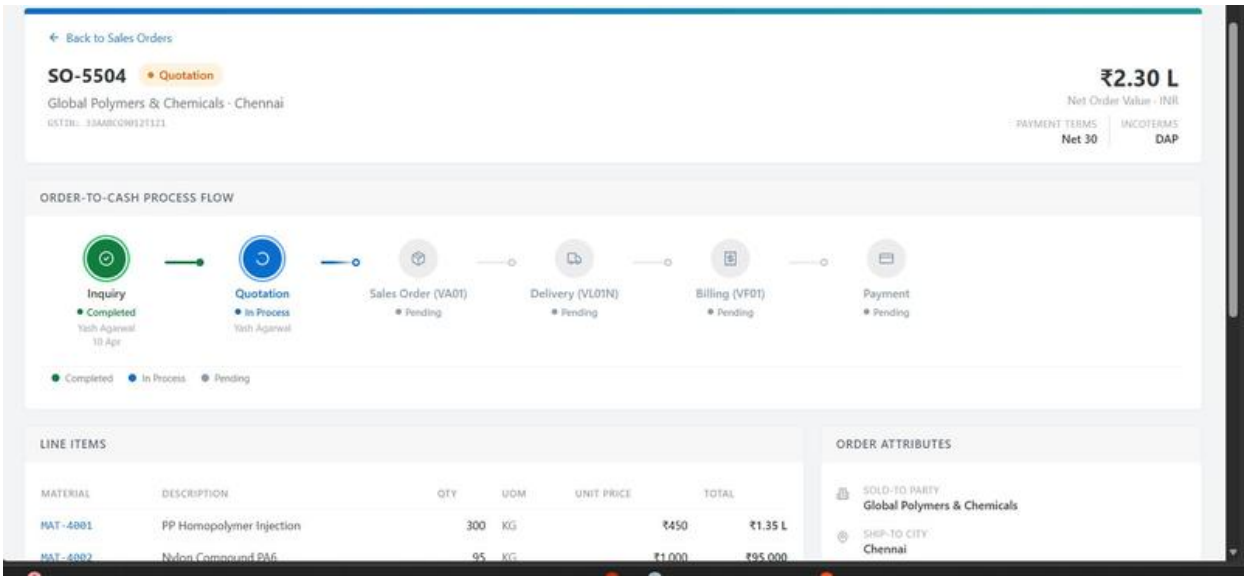


Figure 3 — Order Detail View — Process Flow & Line Items (SO-5504)



— End of Report —

NovaTex Industries Pvt. Ltd. | KIIT SAP BTP Project | Order-to-Cash (O2C)